THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $\frac{3}{4} \div 3$ equals: 1

(a) $2\frac{1}{4}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{8}$

2. Lines in a plane that never meet are called: 1

(a) perpendicular lines **(b)** transversals
(c) intersecting lines **(d)** parallel lines

3. Which of these numbers is divisible by 9? 1

(a) 1827 **(b)** 1829 **(c)** 1830 **(d)** 1828

4. The apex angle of an isosceles triangle is 40°. Each base angle is: 1

(a) 70° **(b)** 90° **(c)** 60° **(d)** 80°

5. The product of two proper fractions is always: 1

(a) smaller than each of the fractions **(b)** larger than each of them
(c) a whole number **(d)** equal to their sum

6. A transversal cuts two parallel lines. One angle is 55°. The corresponding angle is: 1

(a) 45° **(b)** 55° **(c)** 75° **(d)** 65°

7. In the Collatz rule (halve if even, multiply by 3 and add 1 if odd), the number after 7 is: 1

(a) 21 **(b)** 24 **(c)** 23 **(d)** 22

8. The perimeter of a triangle with sides 5 cm, 7 cm and 9 cm (in cm) is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 23

9. The mixed number $2\frac{1}{3}$ written as an improper fraction is: 1

(a) $\frac{5}{3}$ **(b)** $\frac{7}{3}$ **(c)** $\frac{3}{3}$ **(d)** $\frac{7}{2}$

10. A transversal cuts two parallel lines. One interior angle is 48°. Its alternate interior angle is: 1

(a) 38° **(b)** 58° **(c)** 68° **(d)** 48°

11. The sum of the first 6 odd numbers is: 1

(a) 35 **(b)** 36 **(c)** 37 **(d)** 38

12. Two triangles with all three pairs of sides equal are congruent by: 1

(a) ASA **(b)** AAA **(c)** SSS **(d)** SAS

13. A rectangle is $\frac{1}{2}$ m long and $\frac{2}{3}$ m wide. Its area in m^2 is: 1

(a) $\frac{1}{3}$ **(b)** $1\frac{1}{6}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

14. A transversal cuts two parallel lines. Two interior angles on the same side are $62°$ and $x°$. Then $x =$ 1

(a) $108°$ **(b)** $128°$ **(c)** $138°$ **(d)** $118°$

15. A two-digit number has tens digit 7 and units digit 4. Its value is: 1

(a) 76 **(b)** 75 **(c)** 73 **(d)** 74

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Use reverse-and-add on 29 until a palindrome appears. 2

17. Two angles of a triangle are $48°$ and $72°$. Find the third angle and name the triangle by its angles. 2

18. Divide $\frac{3}{4}$ by $\frac{9}{16}$. 2

19. A transversal cuts two parallel lines and one of the angles formed is $70°$. Find the angle vertically opposite to it and the angle forming a linear pair with it. 2

20. Without dividing, test whether 4725 is divisible by 3 and by 9. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A ribbon 6 m long is cut into pieces of $\frac{3}{4}$ m each. How many pieces are made? 3

22. A transversal cuts two parallel lines. An interior angle is $52°$. Find 3
(i) its alternate interior angle and
(ii) the interior angle on the same side as it.

23. Without calculating, decide whether $13 \times 7 + 4$ is even or odd, and give a reason. 3

24. An exterior angle of a triangle is $120°$ and one interior opposite angle is $45°$. Find the other interior opposite angle and the adjacent interior angle. 3

OR

A rectangular card is $\frac{3}{4}$ m long and $\frac{2}{3}$ m wide. Find its area and its perimeter. 3

25. Two angles of a linear pair are in the ratio $1 : 5$. Find them. 3

OR

Apply one step of Kaprekar's routine to 8153 (largest arrangement of the digits minus smallest arrangement). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An angle is 2 times its complement. Find the angle and its complement. 5

OR

Write the Collatz sequence starting from 9. Find the number of steps to reach 1, the largest number reached and how many even numbers occur. 5

27. In $\triangle ABC$, $\angle A = 40°$ and $\angle B = 65°$. Find $\angle C$, classify the triangle by its angles and find the exterior angle at C. 5

OR

A baker has 5 kg of flour. One cake needs $\frac{3}{4}$ kg. How many full cakes can be made and how much flour is left over? 5

END OF QUESTION PAPER **Total: 50 Marks**

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